



BITS Pilani
Pilani Campus



SURVEYING (CE F213)

(Chain Surveying coming under Linear measurement)

Linear Measurement

- In engineering survey, the **linear horizontal distance** between desired points is necessary to be measured to complete a survey.
- The measurement of this linear horizontal distance between two points on the earth surface is known as linear measurement.
- Linear measurement can be achieved by various instruments.

What is the objective of surveying ? What are the principles of surveying ?

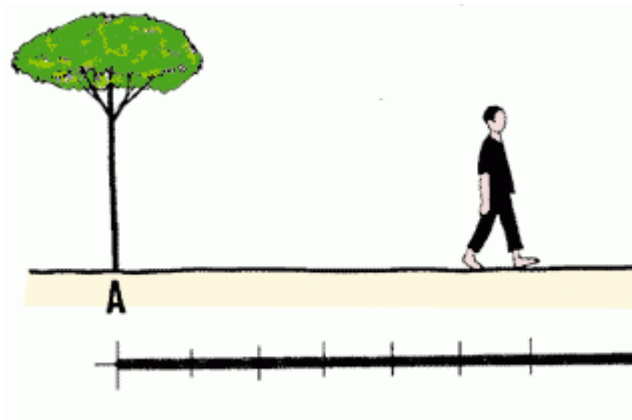
Linear Measurement

The measurement type can be divided as,

1. Direct Method
2. Optical Method
3. E.D.M. Method (Electromagnetic Distance Measuring Method)

Direct method

- ✓ Pacing(Stepping)
- ✓ Passometer (counts the number of steps attached to leg)
- ✓ Pedometer (similar to passometer, length of step or pace can be adjusted)
- ✓ Odometer and speedometer (Odometer is an instrument that measures the number of revolutions of a wheel, Odometer calibrated to give the distance directly is called speedometer(Automobile etc..))
- ✓ Chaining (Chain, tape,....) **[More accurate than the other mentioned methods]**



Pacing



Odometer

Optical Method

The observations can be recorded using **telescope** and the distance can be calculated. **Theodolite** can also be used.

E.D.M. Method (Electromagnetic Distance Measuring Method)

- **Geodimeter**
- **Tellurometer**
- The Decca Navigator and
- **The Lambda Position Fixing System**
- Others

They rely on the measurement of **electromagnetic waves** and measuring even fraction of wavelength by **phase difference technique**. **These even display the distance measured.** They are costly and delicate. Hence for ordinary surveying they are not used commonly.

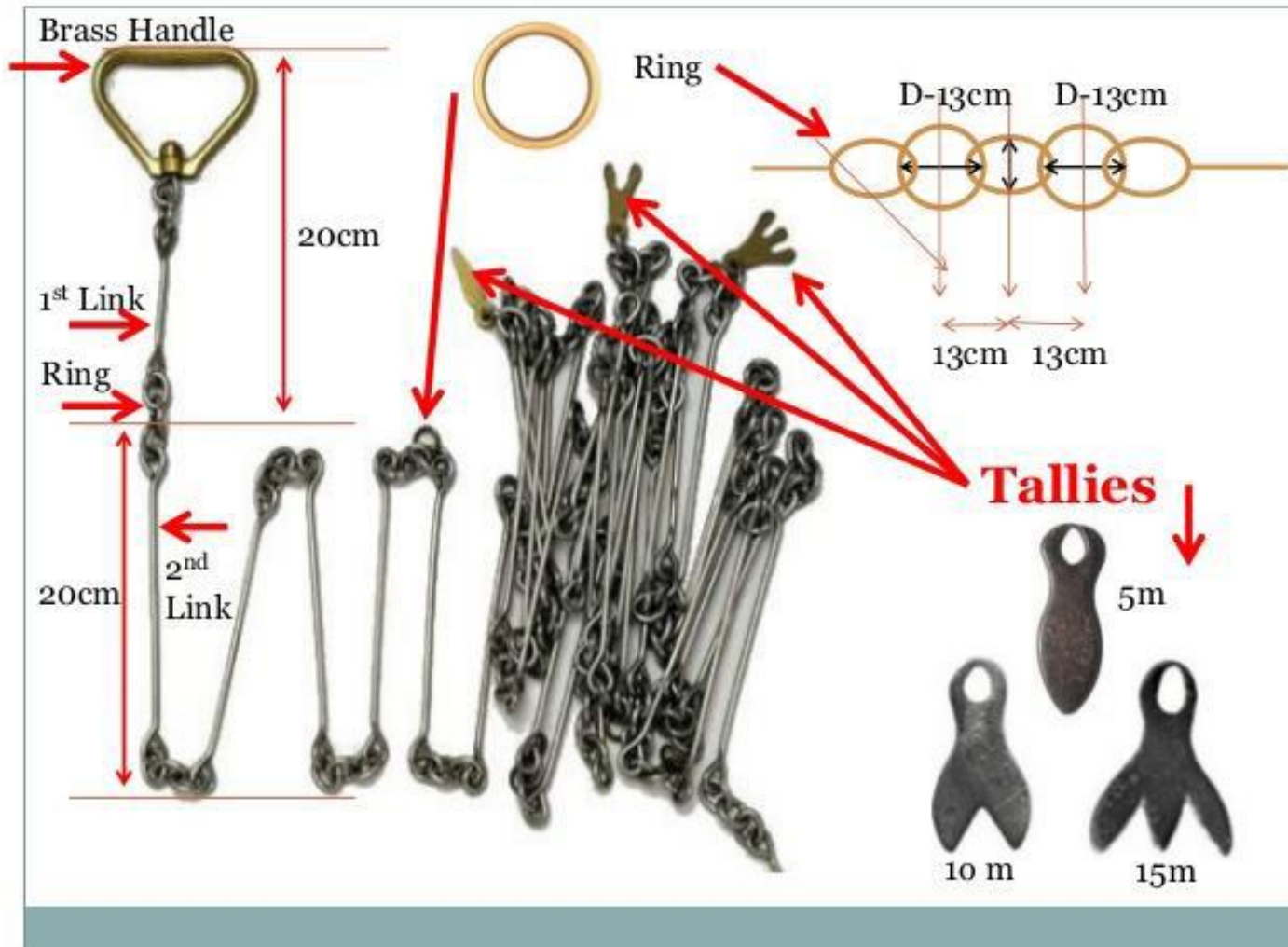
Chaining(chain surveying)

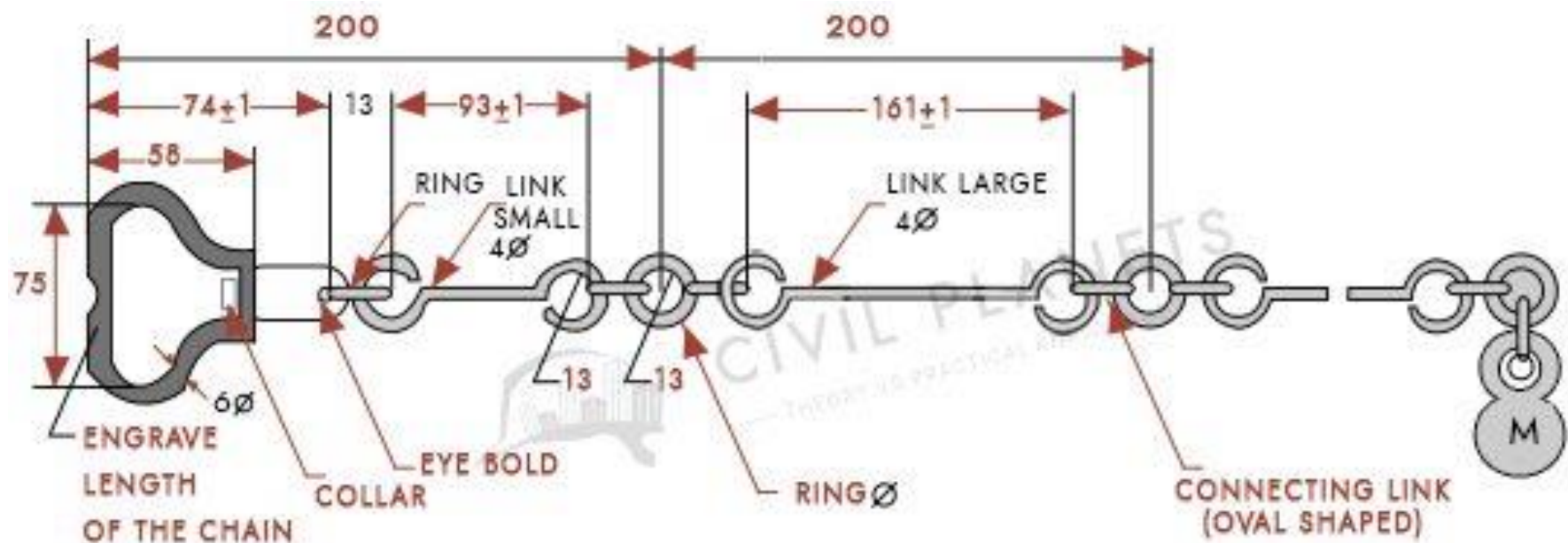
Measuring distance by direct method. The instruments necessary for chain surveying are:

1. Chain
2. Tape
3. Ranging rod
4. Offset rod
5. Arrows (Chain pin)
6. Wooden Pegs and Wooden Hammer
7. Plumb bob
8. Cross staff
9. Clinometer
10. Optical Square
11. Others...

Chain

- Metric chains (20 m and 30m)
- Older chains were in 100 feet (Engineer's), 66 feet (Gunter's or Surveyor's) and 33 feet (Revenue)
- Chains have tallies and rings to identify intermediate values





ALL DIMESIONS IN MILLIMETERS





Tapes

1. Cloth Tape of Linen tape
2. Metallic tape
3. Steel tape
4. Invar tape



All these tapes has their won advantages and dis advantages.

Arrows(Chain Pins or Marking Pins)

Arrow consists of a piece of steel wire about 0.5m long, and are used for marking temporary stations. A piece of coloured cloth, white or red ribbon is usually attached or tied to the end of the arrow to be clearly seen on the field.



Wooden Pegs

Pegs are made of wood 50mm x 50mm(not fixed) and some convenient length(20cm -30cm). They are used for points which are required to be permanently marked, such as intersection points of survey lines. Pegs are drive



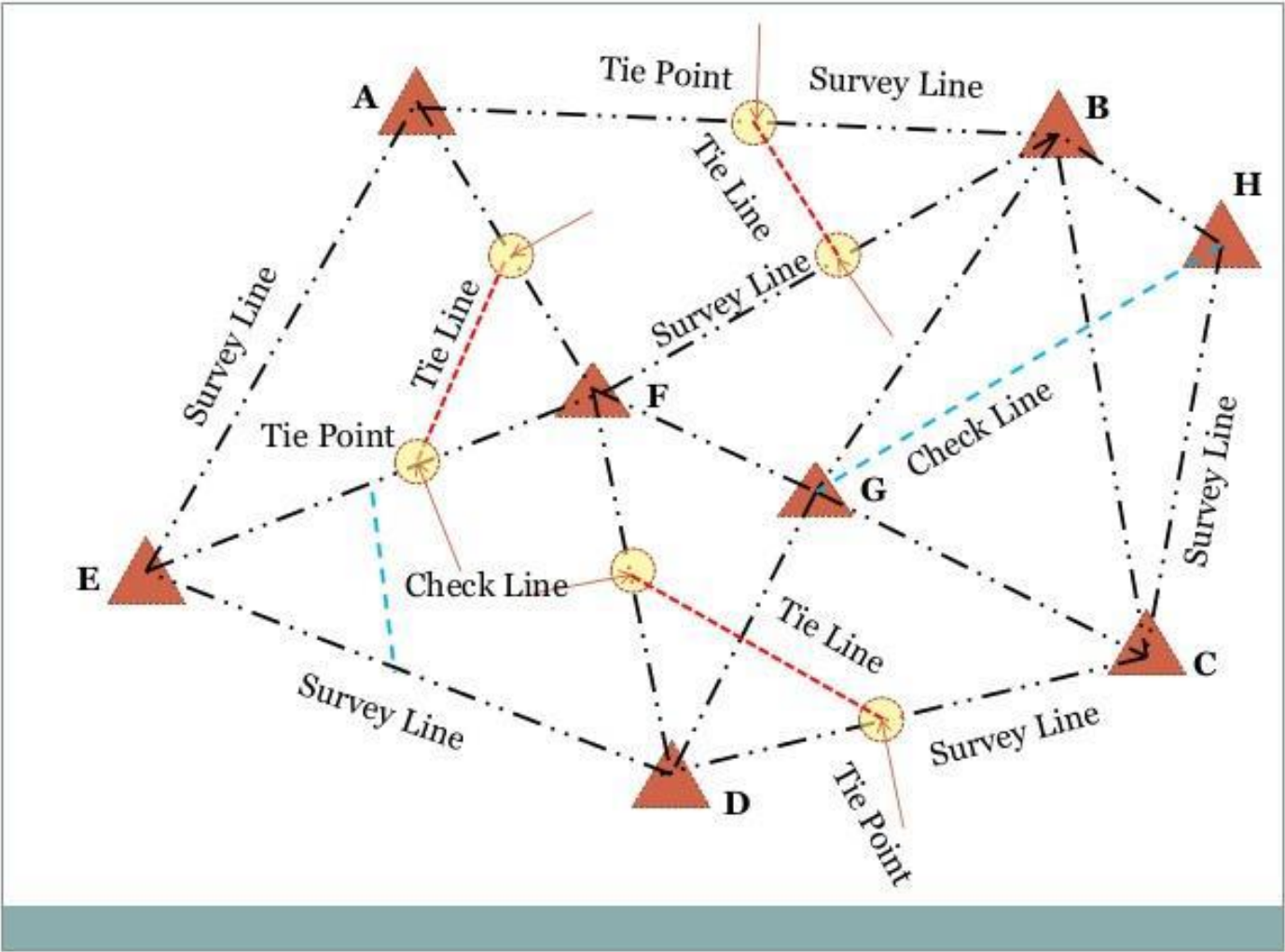
Wooden Hammer: Used to fix the wooden pegs in the ground

Ranging rod

These are poles of circular section 2m, 2.5m or 3m long, painted with characteristic red and white bands which are usually 0.5m long and tipped with a pointed steel shoe to enable them to be driven into the ground. They are used in the measurement of lines with the tape, and for marking any points which need to be seen.



Lines and Stations in Chain Surveying



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- Main station
 - Tie station(Subsidiary station)
 - Main survey line
 - Tie or subsidiary line
 - Base line
 - Check line (Proof line)
 - Offset
 - Chainage

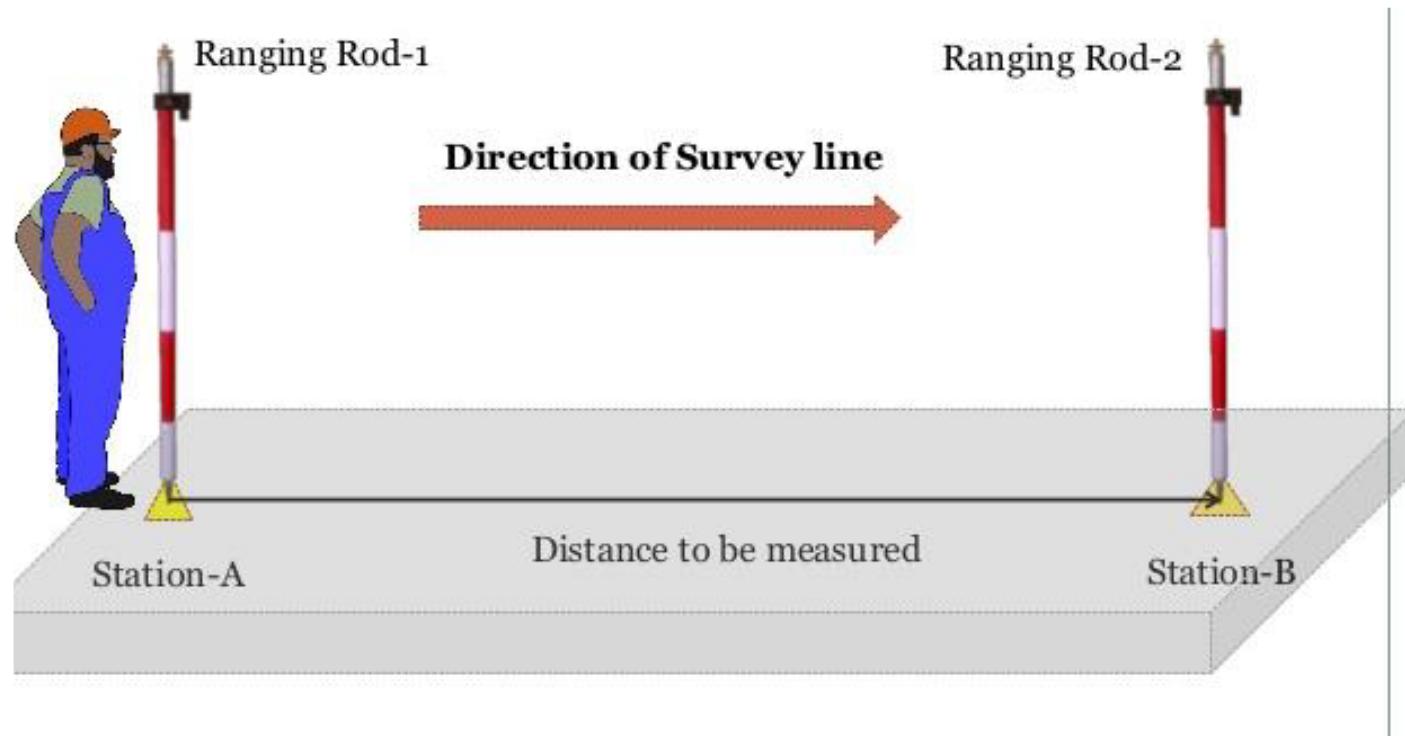
Operations in chain surveying

1. **Chaining:** (Putting the chain in between two main stations)
2. **Ranging:** (If the chain length is less than the length of the line, intermediate stations are to be fixed)
3. **Offsetting:** (when obstacle comes in main chain line or survey line)

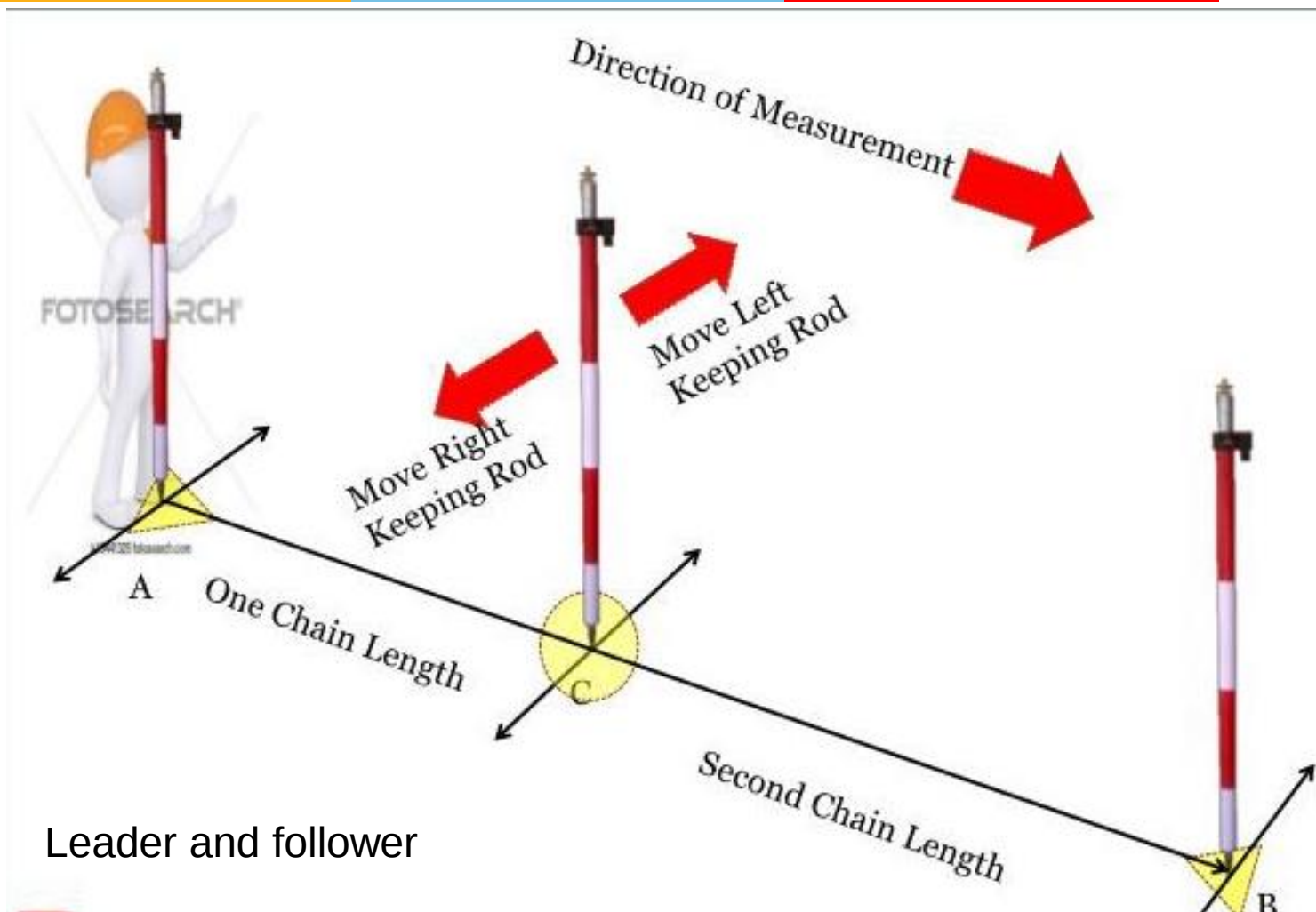
Ranging (fixing intermediate stations by the help of ranging rods when the survey line is lengthier than the chain length)

- **Direct ranging** (if the main stations are inter-visible)
- **Indirect ranging** (Reciprocal ranging) (if the main stations are not inter-visible)

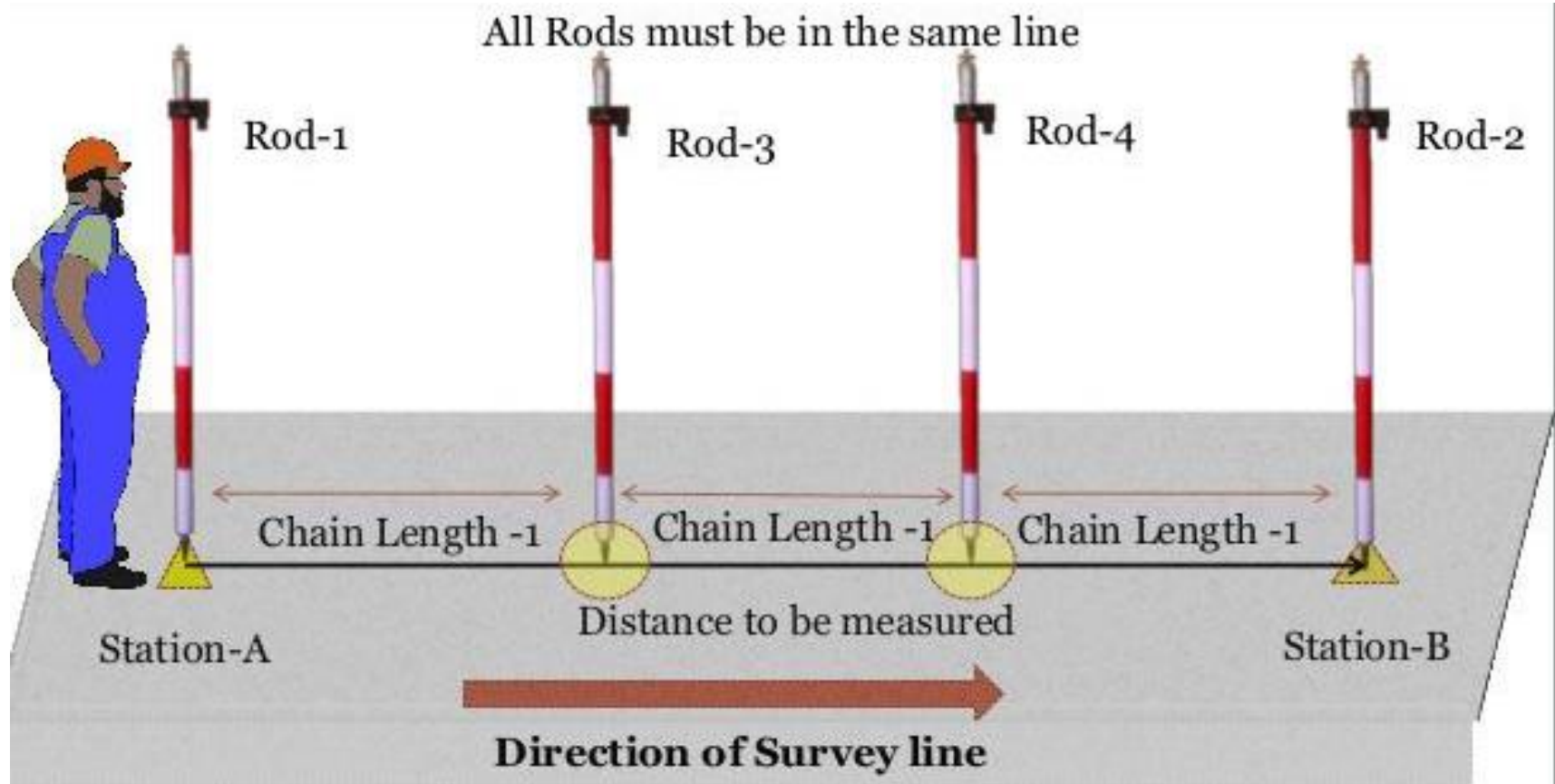
Direct ranging



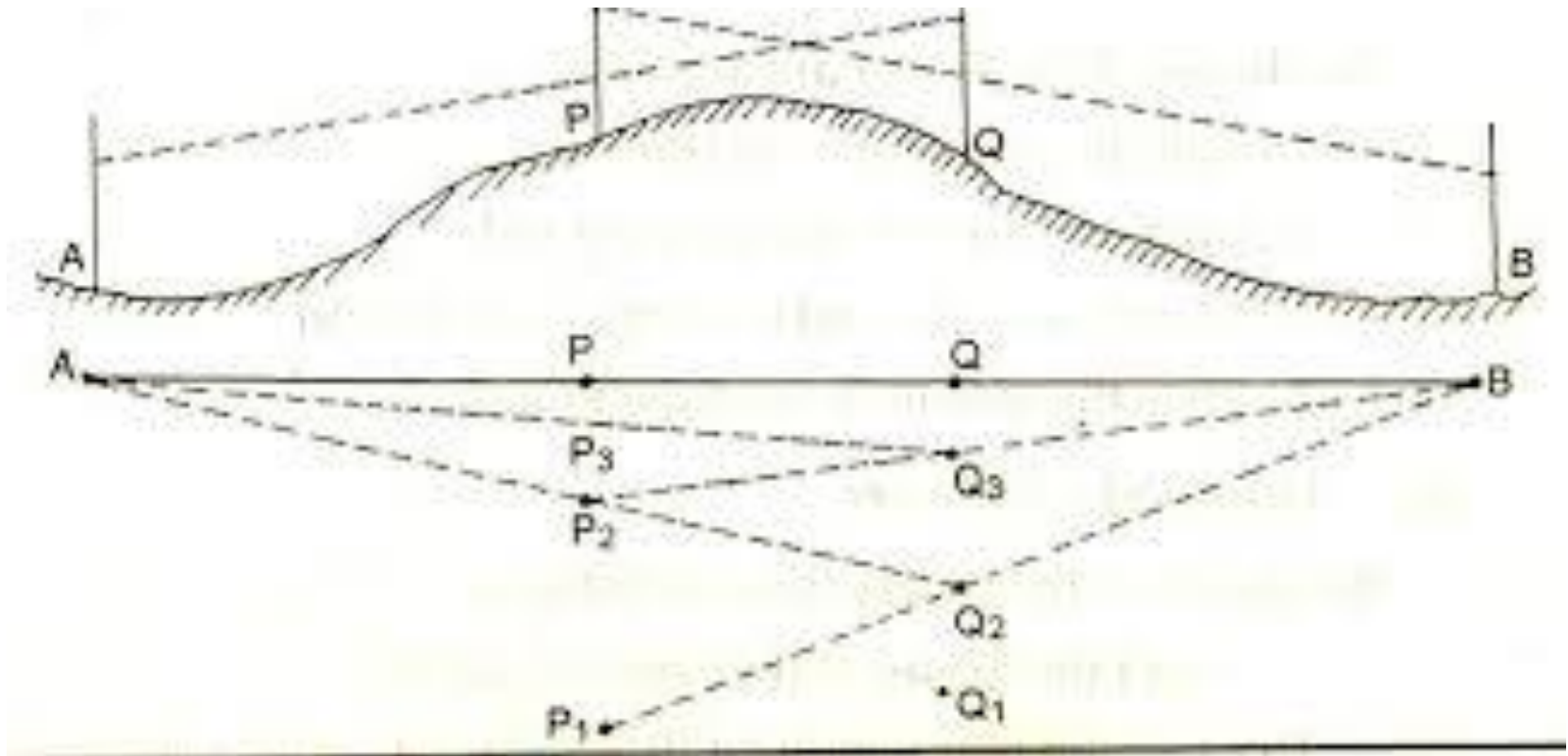
Survey line length is less than the chain length



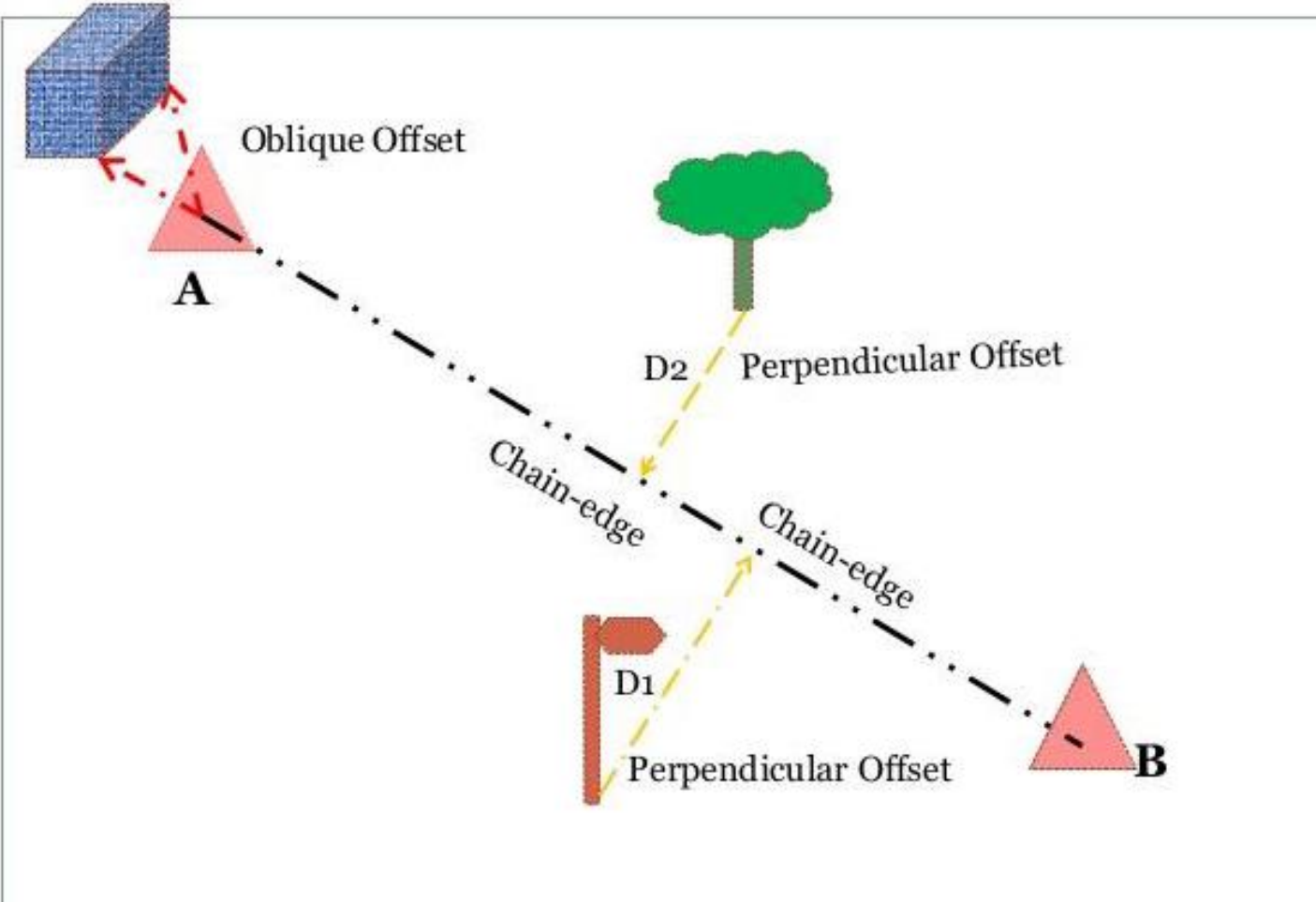
Leader and follower



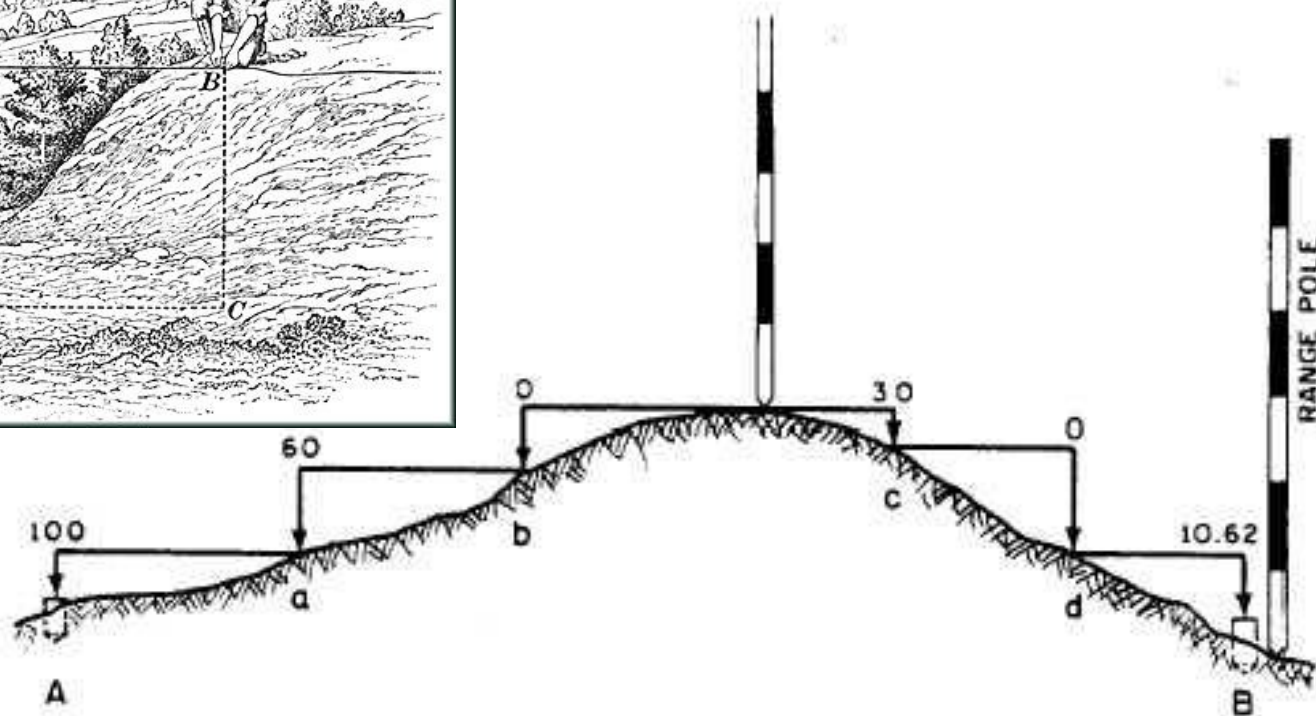
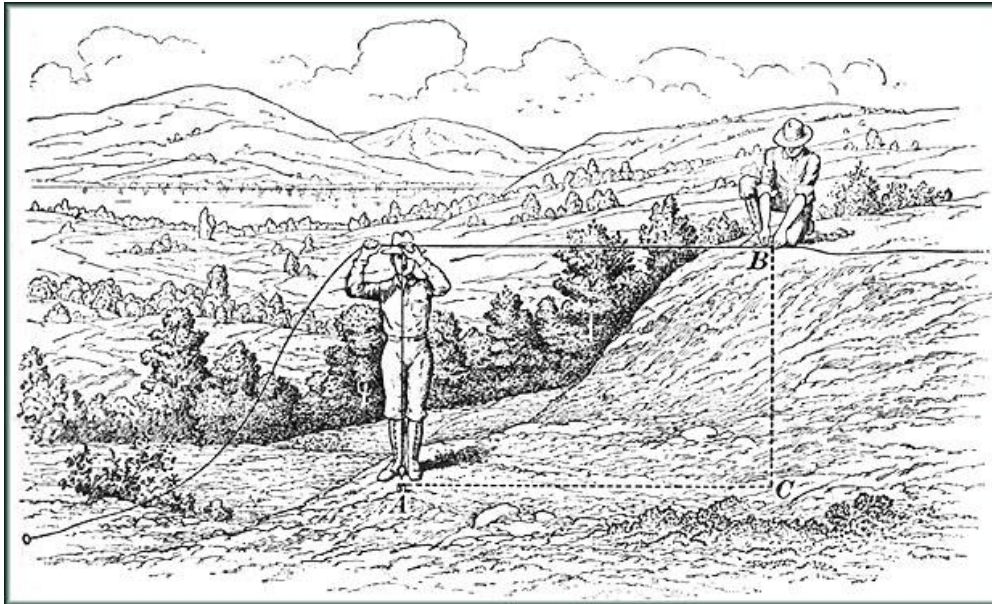
Indirect ranging (Chaining in a mountain and in a valley)



(Ranging in mountain)



Chaining in sloping ground ([Horizontal distance is measured](#))



Errors in surveying (applicable for chain surveying also)

- Surveying is a process that involves observations and measurements with a wide range of electronic, optical and mechanical equipment some of which are very sophisticated.
- Despite the best equipments and methods used, it is still impossible to take observations that are completely free of small variations caused by errors which must be guarded against or their effects corrected.

Gross Errors

- These are referred to mistakes or blunders by either the surveyor or his assistants due to **carelessness or incompetence**.
- On construction sites, mistakes are frequently made by in – experienced Engineers or surveyors who are unfamiliar with the equipment and method they are using.
- These types of errors include miscounting the number of tapes length, wrong booking, sighting wrong target, measuring anticlockwise reading, turning instruments incorrectly, displacement of arrows or station marks etc.
- Gross errors can occur at any stage of survey when observing, booking, computing or plotting and they would have a damaging effect on the results if left uncorrected.
- **Gross errors can be eliminated only by careful methods of observing booking and constantly checking both operations.**

Systematic or Cumulative Errors

- These errors are cumulative in effect and are caused by badly adjusted instrument and the **physical condition at the time of measurement** must be considered in this respect. Expansion of steel, frequently changes in electromagnetic distance (EDM) measuring instrument, etc are just some of these errors.
- Systematic errors have the same magnitude and sign in a series of measurements that are repeated under the same condition, thus contributing negatively or positively to the reading hence, makes the readings shorter or longer.
- This type of error can be **eliminated from a measurement using corrections** (e.g. effect of tension and temperature on steel tape).
- Another method of removing systematic errors is to calibrate the observing equipment and quantify the error allowing corrections to be made to further observations.
- Observational procedures by **re-measuring the quantity with an entirely different method using different instrument** can also be used to eliminate the effect of systematic errors.

Random or Compensating Errors

- Although every precaution may be taken certain **unavoidable errors** always exist in any measurement caused usually by **human limitation** in reading/handling of instruments.
- Random errors **cannot be removed** from observation but methods can be adopted to ensure that they are kept within acceptable limits.
- In order to analyze random errors or variable, statistical principles must be used and in surveying their effects may be reduced by **increasing the number of observations and finding their mean**. It is therefore important to assume those random variables are normally distributed.

Corrections to Linear Measurement in chain surveying,

The following corrections are to be applied to the linear measurements with a chain or a tape,

1. Correction due to faulty chain length
2. Slope correction
3. Standard Length correction of tape
4. Temperature correction
5. Pull correction
6. Sag correction

Errors in measurement due to faulty chain length

Due to excessive use the length of the chain alters gradually. A correction is applied in the measurement taken with the faulty chain.

L- length of original chain

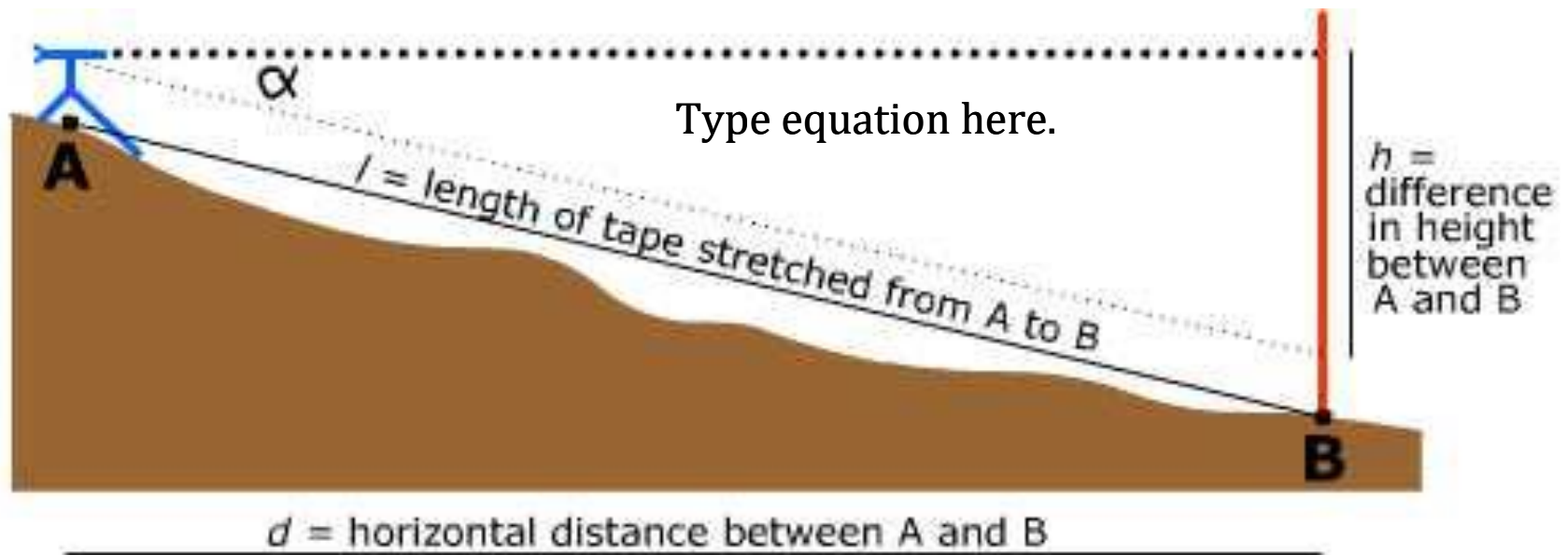
L'- Length of the faulty chain

True length of the line = $\frac{L'}{L}$ X Measured length of the line with the faulty chain

True area = $\left(\frac{L'}{L}\right)^2$ X Measured area

True Volume = $\left(\frac{L'}{L}\right)^3$ X Measured Volume

Slope correction



horizontal distance = $\sqrt{l^2 - h^2}$

Slope correction is $C_s = l - \sqrt{l^2 - h^2}$

$$= l - l\left(1 - \frac{h^2}{l^2}\right)^{\frac{1}{2}}$$

$$= \frac{h^2}{2l} \text{ (neglecting higher order terms of the expansion)}$$

Slope correction is always subtractive

Standard Length correction of tape

The actual length of a tape is calibrated with a standard tape with known length. If the actual length of the tape is not equal to the standard length then a correction is applied in the measured lengths,

$$C_a = \frac{CL}{l}$$

C_a = Correction for actual length

C = Correction to be applied to the tape(+ve or –ve), if the chain length is more than the standard length C will be +ve and, if the chain length is less than the standard length C will be –ve

L = measured length

l = nominal length of the tape



Standard



Too long



Too short

Temperature correction

If the temperatures is different during measurement, than the temperature at which the tape is standardized, then a correction is to be applied

$$C_t = \alpha(T_m - T_0)L$$

T_m = mean temp. during measurement

T_0 = temp. at which the tape is standardized

L = measured length

α = coefficient of thermal expansion (taking same as coefficient of thermal contraction)

= 0.0000035/°C (steel)

= 0.000000122/°C (invar)

if the temp. is higher than the standardized temp, correction is +ve (tape will measure less length than the actual length)- tape expands

if the temp. is lower than the standardized temp, correction is -ve (tape will measure more length than the actual length)

Pull(tension) correction

If the tension during the measurement is different than the tension at which the tape is standardized, a correction is necessary to be applied.

$$C_p = \frac{(P - P_0)L}{AE}$$

P=applied pull during measurement

P_0 = standard pull

A= C/S area of the tape

E= modulus of elasticity of the tape

L =measured length

Pull more than the standard pull tape will expand and it will measure lower length, correction is +ve

Pull less than the standard pull tape will contract and it will measure higher length, correction is -ve

Sag correction

When a tape is stretched between two points, it takes curved form parabolic form (assumed). So a correction is to be applied,

$$C_{sag} = \frac{W^2 L}{24 P^2} = \frac{(wL)^2 L}{24 P^2} \text{ for single span (how to derive this formula)}$$

This correction is always –ve, because due to this higher length is calculated.

Q1: A 30 m tape was standardized at 20⁰ C. What is the correction per tape length if the temperature during measurement was 45⁰ C. Coefficient of thermal expansion = 12 x 10⁻⁶ per degree Centigrade.

Solution

$$C_t = \alpha(T_m - T_0)L = 12 \times 10^{-6} (45-20) \times 30 = 0.009 \text{ m}$$

Q2: Find the sag correction for a tape 30 m long if it was supported at the ends under a pull of 140 N. weight per meter length of the tape was 3 N.

Solution

$$C_{sag} = \frac{W^2 L}{24P^2} = \frac{(wL)^2 L}{24P^2} = 0.516\text{m (Approximate sag correction)}$$

To, find the exact sag correction, we have to use the cubic order equation or to follow the iterative method.

$$L' = L + \frac{w^2 L^3}{24P^2}, \text{ Solving this } L=29.5083, \text{ So Sag correction is } C_{sag} = 30 - 29.5083 = 0.4917\text{m}$$

Q3: 50 m tape was supported at the ends and at the middle. Find the sag correction if the pull during measurement was 150 N and the tape had a cross section of 3 mm². Unit weight of the tape was 78.6 kN/m³.

Solution (one way)

$$C/S \text{ area} = 3 \text{ mm}^2 = 3 \times 10^{-6} \text{ m}^2$$

$$w = 3 \times 10^{-6} \times 78.6 \times 1000 \text{ N/m}^2 = 0.2358 \text{ N/m}$$

Here, there are two spans, So for each span sag correction is calculated and added.

$$\text{For Span-1, } C_{\text{sag}} = \frac{(w)^2 L^3}{24P^2} = \frac{0.2358^2 \times 25^3}{24 \times 150^2} = 0.0016 \text{ m}$$

$$\text{For Span-2, } C_{\text{sag}} = \frac{(w)^2 L^3}{24P^2} = \frac{0.2358^2 \times 25^3}{24 \times 150^2} = 0.0016 \text{ m (same)}$$

$$\text{Total } C_{\text{sag}} = 0.0016 + 0.0016 = 0.0032 \text{ m}$$

(If there are more than one span, sag correction for all spans are found out and added. If there are n no. of same spans, then sag correction for one span is calculated and multiplied by n to find total sag correction)

Solution (the other way)

If there are n no. of same spans,

$C_{\text{sag}} = n \times \frac{(w)^2 L^3}{24P^2}$, where L is the length within one span, n= no. of spans, w is wt. per unit length

$$C_{\text{sag}} = n \times \frac{(w)^2 L^3}{24P^2} = n \times \frac{(w)^2 L^2 L}{24P^2} = \frac{(w^2) \left(\frac{n^2 L^2}{n^2}\right) nL}{24P^2} = \frac{(w^2) (n^2 L^2) nL}{24P^2 n^2} = \frac{(w^2 n^2 L^2) nL}{24P^2 n^2} = \frac{W^2 nL}{24P^2 n^2}$$

nL is total length including all spans, $w^2 n^2 L^2 = W^2$ is square of total weight considering the length in all spans

$$W = wnL = 3 \times 50 \times 78600 / (1000)^2 = 11.79 \text{ N (total load including lengths in all spans)}$$

$$\text{Sag correction} = \frac{W^2 nL}{24P^2 n^2} = 50 (11.79)^2 / (24 \times 2^2 \times 150^2) = 0.0032 \text{ m.}$$

Q4: Before starting the day's work, a 30 m chain was tested and found to be 0.035 m too short. After measuring 880 m, the chain was tested again and it was found to be 0.015 m too long. After measuring another 750 m, the chain was tested at the end of the day and found to be 0.025 m too long. Find the true length measured during the day.

Designated length of chain = 30 m. Actual length at beginning = $30 - 0.035 = 29.965$ m. Actual length after measuring 880 m = $30 + 0.015 = 30.015$ m

Average length during first measurement = $(29.965 + 30.015)/2 = 29.99$

True length = $880 \text{ m} \times 29.99/30 = 879.706$ m

Average length during second measurement = $(30.015 + 30.025)/2 = 30.02$ m

True length = $750 \times 30.02/30 = 750.5$ m

True total length = $879.706 + 750.5 = 1630.206$ m

Derivation of sag correction formula,

$$C_{sag} = \frac{W^2 L}{24P^2} = \frac{(wL)^2 L}{24P^2} \quad (\text{approximated formula, because } L \text{ is horizontal length, but we put curved length as } L \text{ in the formula})$$

(Also to find exact sag correction)

Normal Tension = It is the tension that equalizes correction due to sag and correction due to pull. It is also called **Normal Pull**.

Q5:

Find the normal tension for a tape 30 m long if the calibrating pull was 100 N. The weight of the tape was 15 N, its area of cross section was 4 mm^2 and $E = 200 \text{ GN/m}^2$.

$$C_{sag} = \frac{W^2 L}{24P^2} = \frac{(wL)^2 L}{24P^2} = C_p = \frac{(P - P_0)L}{AE}$$

Solution by trial and error, or bisection method or exact solution of cubic equation,

Normal Pull or Normal tension, $P = 235 \text{ N}$

(Cubic equation can be solved using calculator)

Obstacle in chaining

There are three types of obstacles

1. Chaining free, vision obstructed
2. Chaining obstructed, vision free
3. Chaining and vision both obstructed

1. Chaining free, vision obstructed

Case-I

Slopy ground(mountain),
Indirect ranging or Reciprocal
Ranging method

Case-II

Plane ground (with trees, forest)
Random Line method

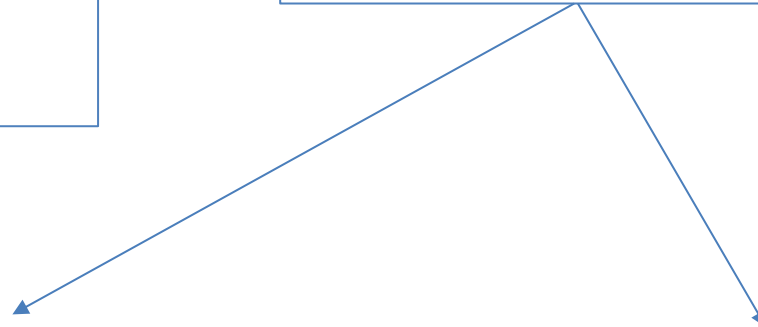
2. Chaining obstructed, vision free

Case-I

Chaining around is free(Pond)

Case-II

Chaining around is not free (River)



Type-i

Chain line Perpendicular to river.

Type-ii

Chain line not Perpendicular to river.

3. Chaining and vision both obstructed

(Building)



Thank You